

Orhan AYDIN

Mechatronics Engineer | AI & Robotics

+44 7538 119910

orhanaydinmechatronic@gmail.com

linkedin.com/in/orhan-aydin/

Bournemouth, United Kingdom

https://github.com/orhanaydinn

SUMMARY

Mechatronics Engineer and MSc Data Science & Artificial Intelligence graduate focused on robotics, embodied AI, robot manipulation, computer vision, and robot learning. Experienced in developing end-to-end robotic systems, including mechanical design, 3D printing, embedded control, perception, kinematics, simulation, and LLM-enabled decision-making. Built a custom desktop humanoid robot with a 5-DOF arm, gripper, vision-based object detection, natural-language command execution, and safety validation layers. Skilled in Python, C++, PyTorch, OpenCV, MuJoCo, SolidWorks, Arduino, and real-world robotic prototyping.

EDUCATION

MSc Data Science and Artificial Intelligence with Distinction	01/2025 – 07/2026
Weighted Average: 75.56%	
Bournemouth University – Master of Science	United Kingdom
BSc Mechatronic Engineering	09/2016– 09/2022
GPA: 2.92 / 4.00	
Erciyes University – Bachelor of Science	Turkey

PROJECTS

- **Design and Development of an LLM-Enabled Humanoid Robot (MSc Thesis)**
 - Designed and developed a custom fixed-base humanoid robot from scratch, including 3D-printed mechanical structure, electronics integration, embedded control, and full hardware–software integration.
 - Built a 5-DOF robotic arm with gripper, camera-based object detection, depth perception, inverse kinematics, voice interaction, and low-level robot control for real-world pick-and-place behaviours.
 - Developed a vision-language control pipeline using JSON-based robot tool calling with Planner and Controller validation layers to repair, reject, clarify, or safely execute LLM-generated actions.
- **Robot Learning for 5-DOF Humanoid Arm Manipulation using MuJoCo**
 - Developed a MuJoCo simulation environment for a custom 5-DOF humanoid robotic arm and gripper.
 - Collected multi-height demonstration datasets for ready-to-pregrasp and pregrasp-to-final grasp manipulation stages, including object pose, target pose, joint angles, gripper state, IK error, and task success data.
 - Trained Behavior Cloning models and prepared Reinforcement Learning fine-tuning experiments to compare BC-only and RL-fine-tuned policies using success rate, final distance error, object displacement, and trajectory quality.
- **Pre-training and Fine-Tuning of a Custom Large Language Model for Human-Robot Interaction**
 - Pre-trained a custom 1B-parameter LLaMA-style language model from scratch using large-scale text datasets, a custom tokenizer, and a transformer-based architecture.
 - Built an end-to-end training pipeline covering data preprocessing, tokenizer training, checkpointed pre-training, supervised fine-tuning, and evaluation.
 - Fine-tuned the model on custom human–robot interaction data to generate structured JSON robot tool calls such as search_object, pick_object, place_object, and handover_object.
 - Integrated the model into the humanoid robot pipeline for natural-language command interpretation and high-level robot action selection.

EXPERIENCE

Mold Specialist Engineer

08/2023 – 08/2024

Yorglass

Turkey

- Designed glass press machine molds using SolidWorks and AutoCAD for automated production environments.
- Improved machine production speed from 10 rpm to 12.7 rpm through mold and process optimization.
- Designed customized robotic gripper solutions, reducing robot-related production losses from 25% to 5% per day.
- Collaborated with production and maintenance teams to improve mechanical reliability and automation performance.

Maintenance Engineer

10/2022 – 08/2023

Yorglass

Turkey

- Supported the installation and maintenance of automated production lines for glass press machines.
- Performed mechanical and electrical maintenance on industrial production systems.
- Implemented robot transfer solutions using image processing to improve handling and production flow.

Voluntary Project Engineer

10/2019 – 05/2022

Mem R&D

Turkey

- Mentored student teams in national competitions such as Teknofest and TUBITAK. Contributed to embedded system design, project planning, and technical coaching. Supported preparation for presentations and prototype development.

CERTIFICATIONS & TRAINING

AI & Robotics

- Deep Learning with PyTorch and Keras – BTK Akademi
- Python for Data Science with TensorFlow – BTK Akademi
- Machine Learning Applications with Python – BTK Akademi
- Python & OpenCV for Image Processing – Udemy
- Robotic Programming – Baskent University
- PCB Design with Altium Designer – Udemy

Software Development

- Full Stack Web Developer Program (432 hours) – Cosmios Akademi
- C# Programming – BTK Akademi

SKILLS

Robotics & Simulation

- Robot Manipulation, Robot Learning, Behavior Cloning, Reinforcement Learning, MuJoCo, Sim-to-Real Learning, Inverse Kinematics, Forward Kinematics, Motion Planning, Robotic Grippers, Sensor Integration

Computer Vision & Embodied AI

- OpenCV, YOLO, Depth Perception, Object Detection, Visual Servoing, Vision-Language Robot Control, LLM Tool Calling, Human-Robot Interaction

AI & Machine Learning

- PyTorch, TensorFlow, Keras, Transformers, Hugging Face, PEFT/LoRA, Machine Learning, Deep Learning, Computer Vision, Large Language Models

Programming & Engineering Tools

- Python, C++, SQL, Git, Docker, Streamlit

Embedded Systems & Hardware

- Arduino, STM32, ESP32, Raspberry Pi, Microcontrollers, Embedded Control, Electronics Integration

CAD & Mechanical Design

- SolidWorks, AutoCAD, Altium Designer, 3D Printing, Mechanical Design, Prototyping

LANGUAGE

- English – Upper-Intermediate (B2)
- Turkish – Native